

MINI PROJECT -1

Smart Sales Performance Analytics & Forecasting Dashboard

◆ Problem Statement

Organizations collect large volumes of sales data but struggle to convert it into actionable insights. Management needs a unified dashboard to **analyze sales trends, detect patterns, identify top-performing regions/products, and forecast future sales.**

🎯 Objectives

- Analyze historical sales data
 - Identify seasonal trends and sales patterns
 - Compare region-wise and product-wise performance
 - Forecast future sales using ML
 - Provide an interactive dashboard using **Streamlit**
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📋 Tasks to be Performed

1. Data ingestion (CSV)
 2. Data cleaning & preprocessing
 3. Feature engineering (Revenue, Month, Quarter)
 4. Exploratory Data Analysis (EDA)
 5. Pattern detection (seasonality, growth trends)
 6. Sales forecasting using ML
 7. Interactive dashboard creation
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📊 Visualizations & Patterns

- Bar Chart → Revenue by Region
- Line Chart → Monthly Sales Trend
- Heatmap → Product vs Region Sales
- Pie Chart → Category Contribution
- Forecast Plot → Actual vs Predicted Sales

🔧 Technologies Used

- Python
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn
- Streamlit

Solution Approach

ML Model

- Linear Regression / Random Forest for forecasting

Pattern Insights

- Seasonal sales spikes
- Region dominance
- High-performing product categories